

BASIC PRINCIPLES OF MEDICINE 1: **FOUNDATIONS TO MEDICINE**

LECTURE FTM 8 **DNA Replication**

Cristofre Martin
cmartin@sgu.edu

Copyright

All year 1 courses materials, whether in print or online, are protected by copyright. The work, or parts of it, may not be copied, distributed or published in any form, printed, electronic or otherwise.

As an exception, students enrolled in year 1 of St. George's University School of Medicine and their faculty are permitted to make electronic or print copies of all downloadable files for personal and classroom use only, provided that no alterations to the documents are made and that the copyright statement is maintained in all copies.

View only files, such as lecture recordings, are explicitly excluded from download and creating copies of these recordings by students and other users are strictly illegal.

The author of this document has made the best effort to observe current copyright law and the copyright policy of St George's University. Users of this document identifying potential violations of these regulations are asked to bring their concern to the attention of the author.

Objectives: DLA: Structure and Function of Nucleic Acids

- Before lecture ‘DNA replication’

SOM.MK.I.BPM1.1.FTM.3.BCHM.0020	Distinguish between purines, pyrimidines; ribonucleosides, deoxyribonucleosides; ribonucleotides and deoxyribonucleotides.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0021	Describe the phosphodiester bond
SOM.MK.I.BPM1.1.FTM.3.BCHM.0022	Describe the three important features of the DNA double helix (complementary, antiparallel, and forces stabilizing the helix).
SOM.MK.I.BPM1.1.FTM.3.BCHM.0023	Discuss the physicochemical properties of DNA
SOM.MK.I.BPM1.1.FTM.3.BCHM.0024	Describe the basic structures and roles of the major classes of RNA (mRNA, tRNA, rRNA, snRNA, miRNA)
SOM.MK.I.BPM1.1.FTM.3.BCHM.0025	Predict the sequence of the complementary strand if the sequence of one of the DNA strands is provided.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0026	Analyze the nucleotide composition of double stranded DNA using the Chargaff's rule

Objectives: DLA: DNA Packaging

- Before lecture ‘DNA replication’

SOM.MK.I.BPM1.1.FTM.3.BCHM.0028	Differentiate between a supercoiled and a relaxed DNA molecule
SOM.MK.I.BPM1.1.FTM.3.BCHM.0029	State the reasons why supercoiling of DNA molecules is important.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0030	Compare and contrast DNA packaging in prokaryotes and eukaryotes.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0031	Describe the structure of chromatin and histone proteins.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0032	Explain how dsDNA and histones associate to form the nucleosome.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0033	Describe the different structural conformations of DNA packaging in eukaryotes (10 nm Fiber , 30 nm fiber and metaphase chromosomes).
SOM.MK.I.BPM1.1.FTM.3.BCHM.0034	Describe how histone modification can influence transcriptional activity

Objectives: DLA: DNA Replication and Telomerase

- Before lecture ‘DNA replication’

SOM.MK.I.BPM1.1.FTM.3.BCHM.0048	Describe the role of telomeres and telomerase in eukaryotic DNA replication
---------------------------------	---

Objectives: DNA Replication

SOM.MK.I.BPM1.1.FTM.3.BCHM.0035	Describe semi-conservative replication of DNA
SOM.MK.I.BPM1.1.FTM.3.BCHM.0036	Compare and contrast the similarities and differences between DNA Pol I and DNA Pol III of E. coli
SOM.MK.I.BPM1.1.FTM.3.BCHM.0037	Describe how DNA is synthesized from its 5' to 3' end
SOM.MK.I.BPM1.1.FTM.3.BCHM.0038	Explain the significance of the origin of replication during the process of DNA synthesis.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0039	Explain the problems associated with unwinding of the DNA double helix
SOM.MK.I.BPM1.1.FTM.3.BCHM.0040	Describe the role of helicase enzymes in unwinding DNA prior to its replication
SOM.MK.I.BPM1.1.FTM.3.BCHM.0041	Describe the role of single-strand binding protein & topoisomerases in DNA replication
SOM.MK.I.BPM1.1.FTM.3.BCHM.0042	Outline the order of events from RNA priming to completed DNA during replication in E.coli.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0043	Distinguish between synthesis of leading and lagging strands of DNA. Identify the significance of DNA ligase in lagging strand synthesis
SOM.MK.I.BPM1.1.FTM.3.BCHM.0044	Describe the significance of the various proteins and enzymes required for replication and predict the effects of mutations of these proteins and enzymes on DNA replication
SOM.MK.I.BPM1.1.FTM.3.BCHM.0045	Evaluate the proofreading activity of DNA polymerase
SOM.MK.I.BPM1.1.FTM.3.BCHM.0046	Differentiate between DNA replication in eukaryotes and prokaryotes
SOM.MK.I.BPM1.1.FTM.3.BCHM.0047	Outline the mechanisms of action of drugs that interfere with DNA replication

Pre-lecture Reading

View Panopto videos:

Nucleic Acids

DNA Packaging

DNA Replication

Telomerase

Recommended Reading

- Supplemental Study notes for Nucleic Acids, DNA packaging, and DNA replication:
- Lippincott's Biochemistry 9th Edition;
<https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=260214405&bookid=3402>; Chapter 30: DNA Structure, Replication, and Repair
- Lippincott's Biochemistry Chapter 30: [30. DNA Structure, Replication, and Repair | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library](#)

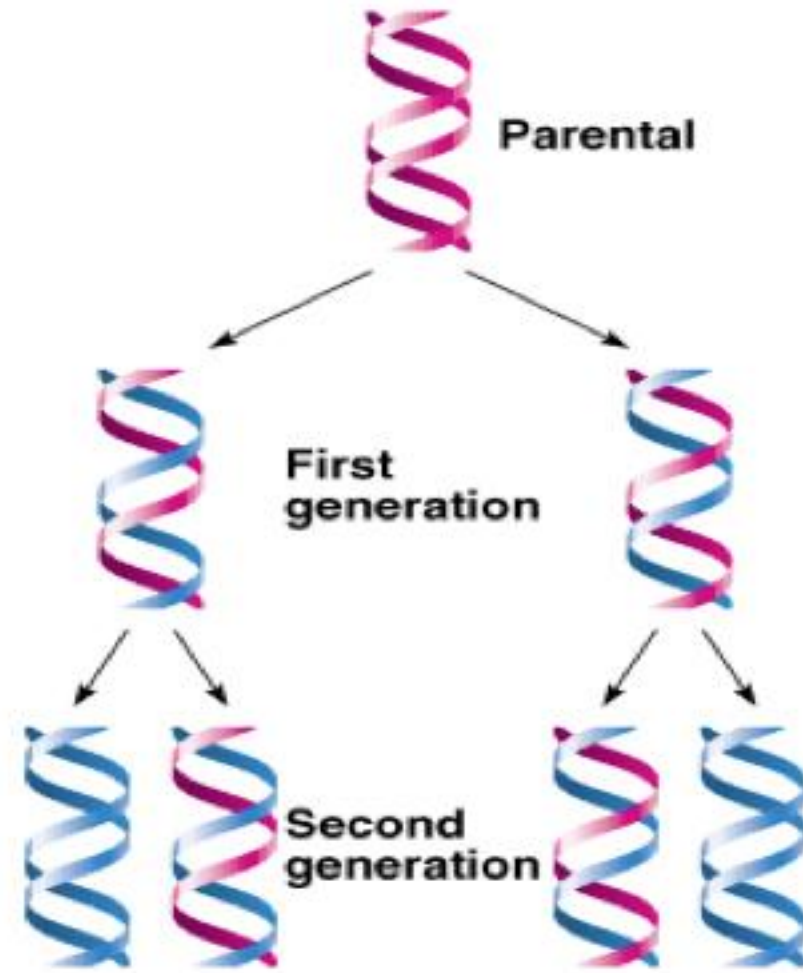
DNA Replication

-the co-ordinated duplication of DNA during the process of cell division

-provides the faithful transmission of hereditary information

-DNA replication in bacteria takes approx. 40 minutes, but in eukaryotes can vary from 1.4 hours in yeast to 24 hours in cultured animal cells

-DNA replication is *semi-conservative*

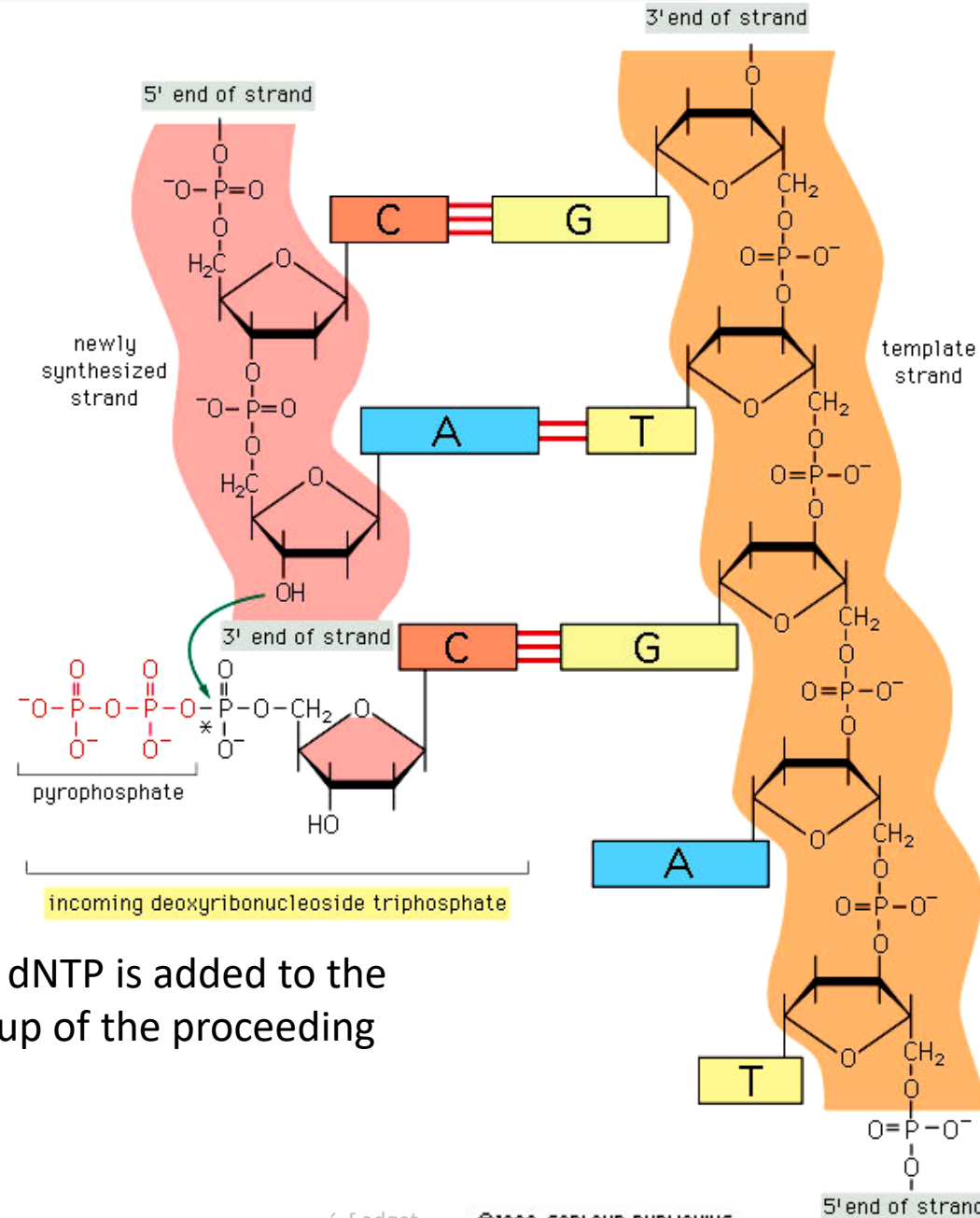
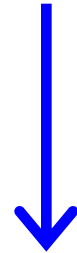


The following components are required for the synthesis of DNA:

- 1) All four dNTPs (deoxynucleoside triphosphate)
-building blocks of the DNA molecule
- 2) A fragment of DNA to act as template.
- 3) DNA polymerase.
- 4) Magnesium ions (Mg^{2+}).
-required for DNA polymerase activity.
- 5) A primer providing a free 3' -OH group.

Chemistry of Chain Elongation

Synthesis in the
5'-3' direction

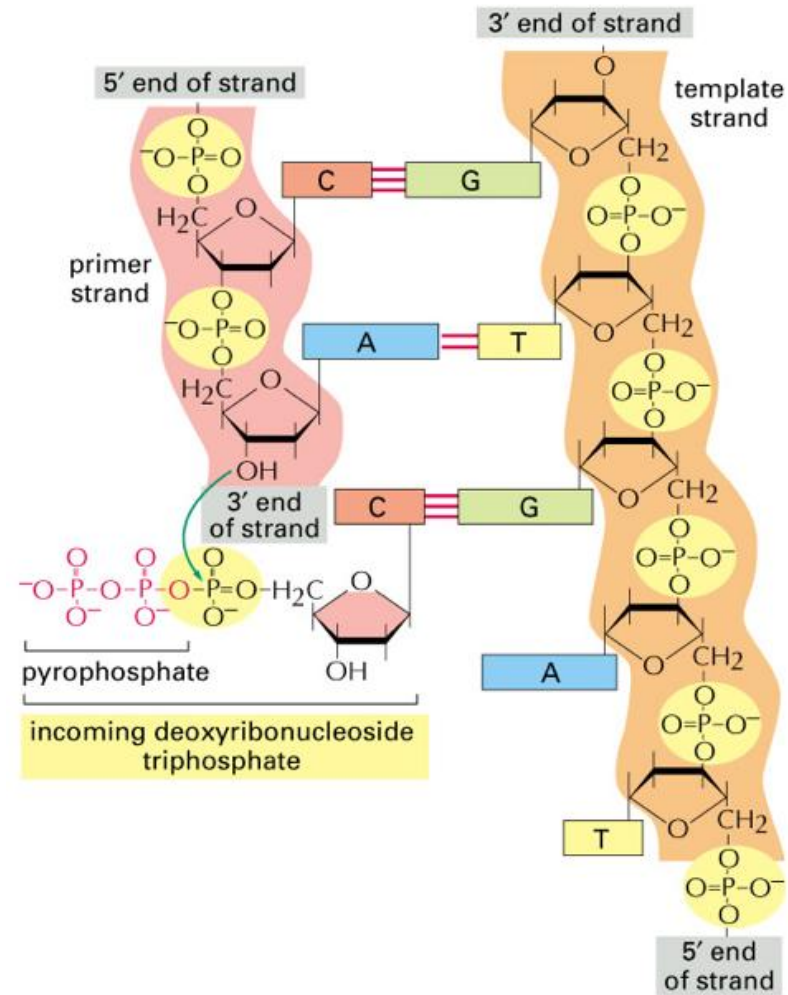


Each new dNTP is added to the
3' OH group of the proceeding
dNTP

DNA chain elongation catalyzed by DNA polymerase

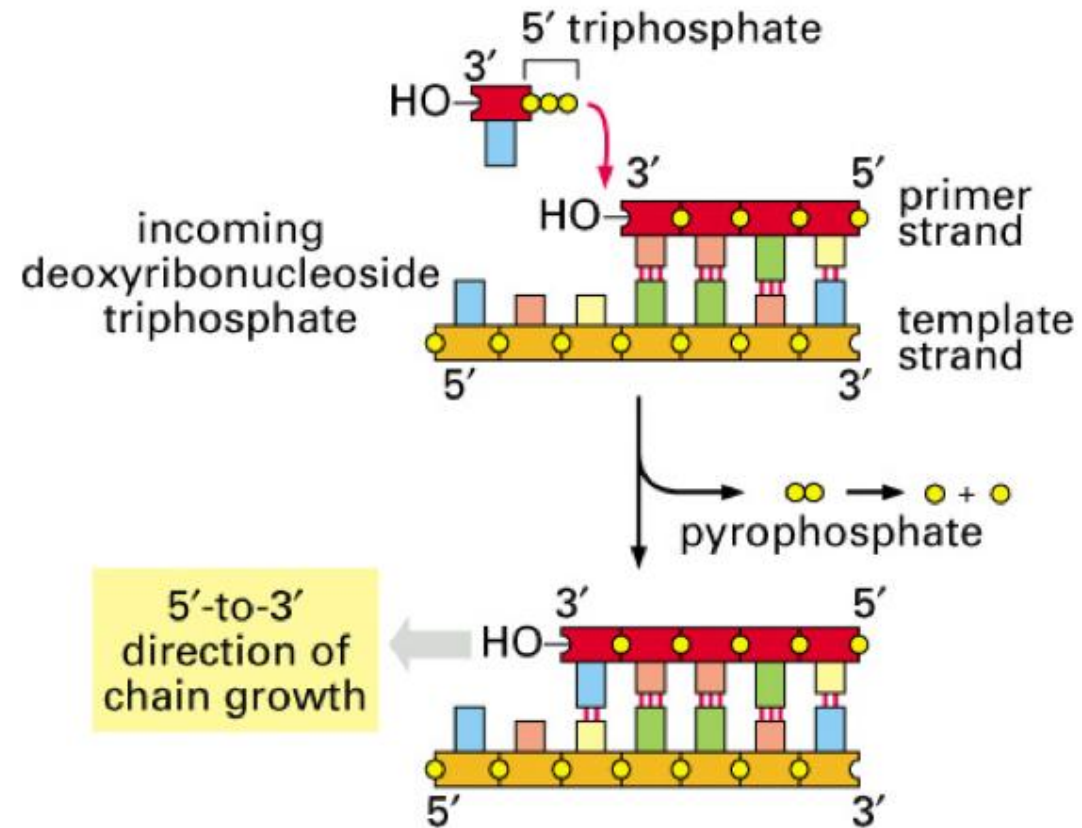
Roles and properties of DNA polymerase:

- 1) DNA polymerase catalyzes the formation of a phosphodiester bond between the 3' OH group of the deoxyribose on the last nucleotide and the 5'-phosphate of the dNTP precursor.
- 2) The deoxynucleoside 5' triphosphate provides the energy source for the reaction.

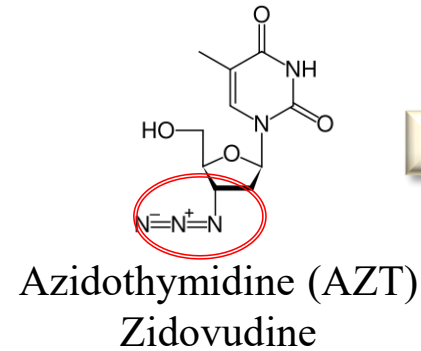
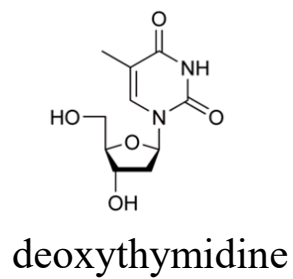


3) At each step in the lengthening the new DNA chain, DNA polymerase finds the correct precursor dNTP that can form a complementary base pair with the nucleotide on the template strand of the DNA.

Pyrophosphate is liberated as a result of this incorporating a single base.



Anti-viral Nucleoside analogues – Viral reverse transcriptase inhibitors



Host cellular kinases



Utilized by the viral reverse transcriptase



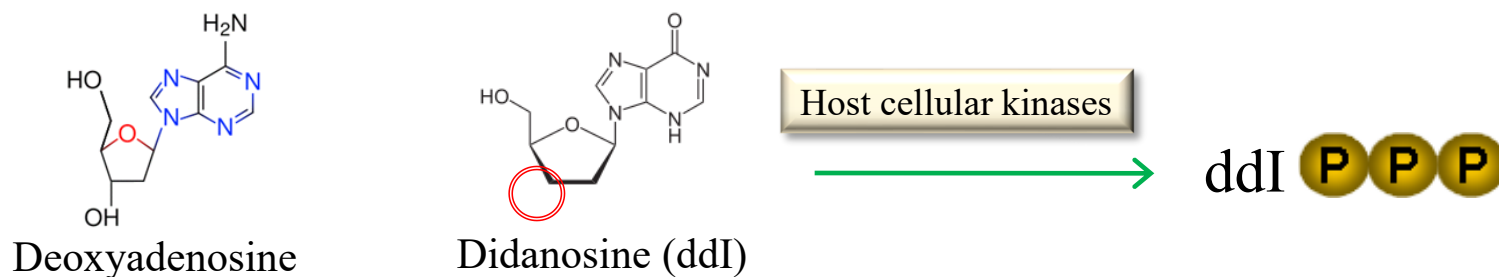
Upon incorporation into the ds DNA chain
termination occurs due to lack of 3'OH

ds = double stranded

Zidovudine can:-

- Prolong life in HIV infected individuals
- Reduce mother to baby transmission by more than 20%

Anti-viral Nucleoside analogues – Viral reverse transcriptase inhibitors



Utilised by the viral reverse transcriptase

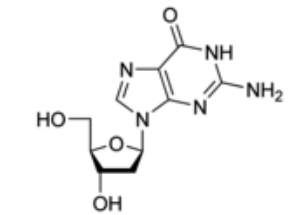


Upon incorporation into the ds DNA chain
termination occurs due to lack of 3'OH

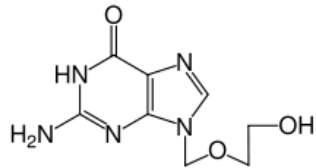
Specificity for infected cells lies in the fact that these drugs have a 100-300 fold greater affinity for incorporation by reverse transcriptase than eukaryotic DNA polymerase.

Used in the treatment of mainly HIV in combination with other anti-virals

Anti-viral Nucleoside analogues – Viral DNA polymerase inhibitor



Deoxyguanosine



Acyclovir

Viral kinase



Acyclovir



*Converted to the mono-phosphate
by viral thymidine kinase*

Host cellular
kinases



Acyclovir



High specificity for
herpes simplex &
varicella zoster.



*Herpes simplex can causes
conjunctivitis and coldsores*

Viral thymidine kinase = much more effective in catalyzing
1st phosphorylation event than host thymidine kinase

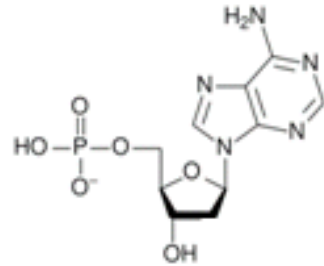


Only infected cells possess the viral kinase
∴ Acyclovir is only activated in infected cells.

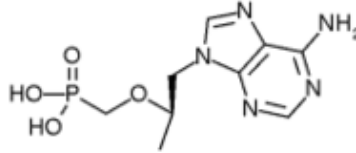


Causes chain termination due to lack of 3'OH
30 x more potent against viral DNA polymerase than host.

Anti-viral Nucleotide Base analogues



Deoxyadenosine
monophosphate



Tenofovir



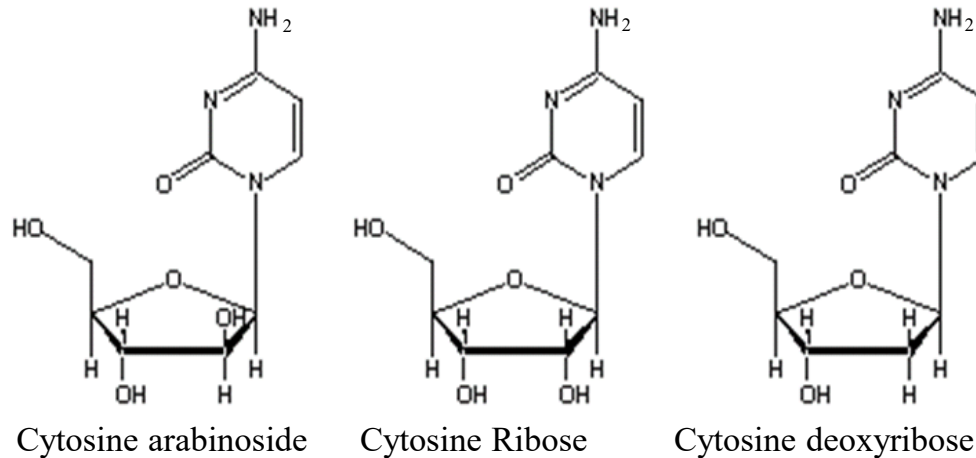
Host cellular
kinases

Tenofovir diphosphate competes with its natural nucleotide counterpart, deoxyadenosine 5'-triphosphate, for incorporation into newly forming **HIV** DNA.

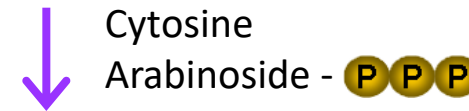


Once successfully incorporated, termination of the elongating DNA chain ensues, and DNA synthesis is interrupted.

Cytosine arabinoside (araC) = cytosine nucleoside analogue



In cells it is rapidly converted to cytosine arabinoside triphosphate



Acts as a substrate for several human DNA Pols

Ribose sugar replaced by arabinose

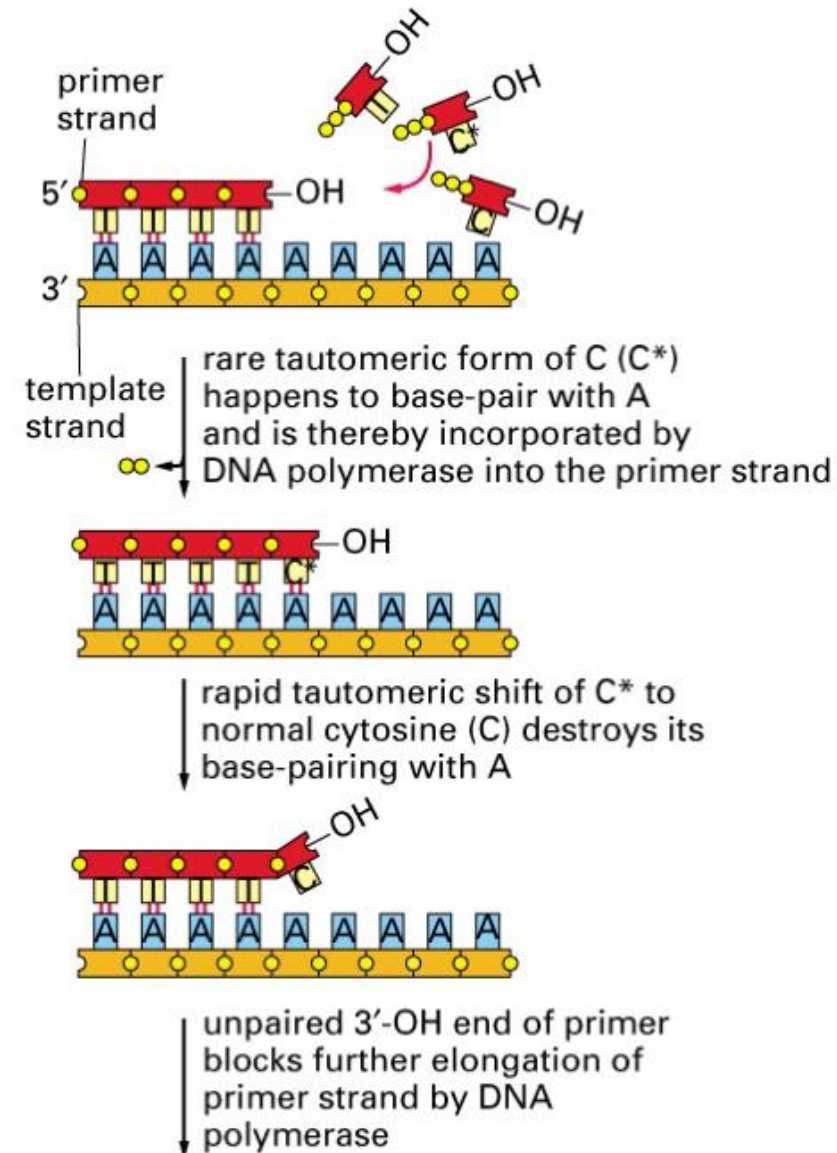
- ❖ AraC has a 3' OH group to act as an acceptor for further chain elongation. However the OH gp at posn., 2 prevents polymerase adding the next dNTP due to stereo restraints → chain termination.

There is also an adenosine nucleoside analogue called adenosine arabinoside that acts in a similar fashion.

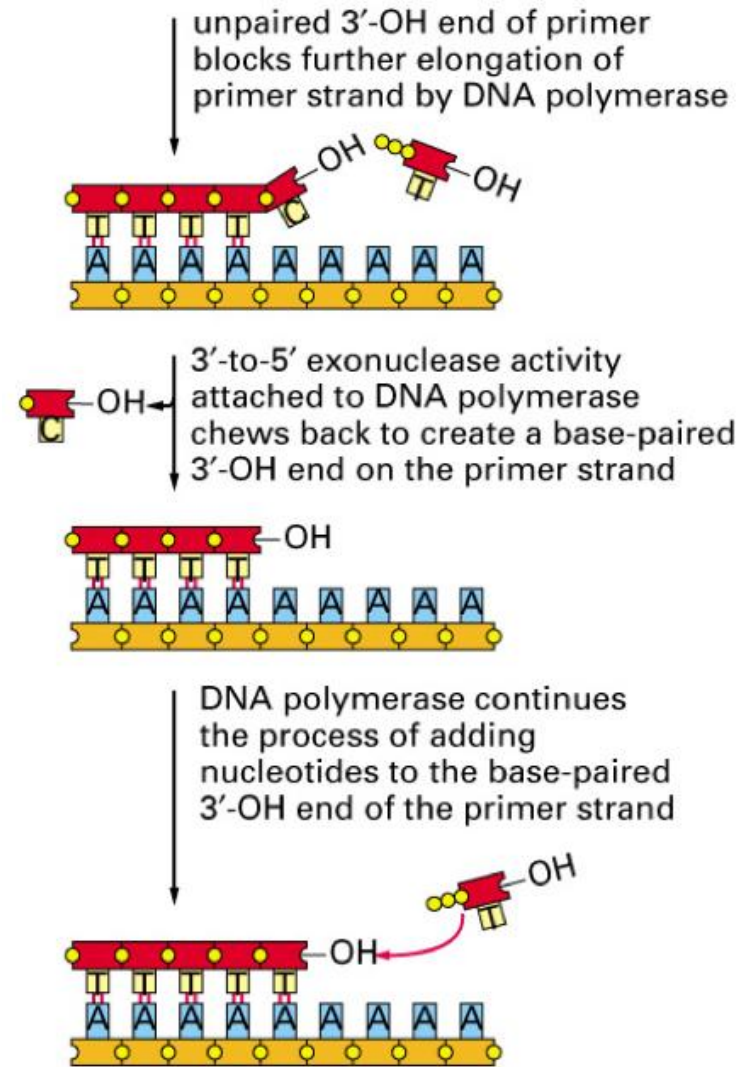
Important in the therapy of leukaemias, in particular acute myeloid Leukaemia and Lymphoma <http://www.touchbriefings.com/pdf/2812/Momparler.pdf>

Repair of base misincorporation by DNA polymerase

- 3) Occasionally DNA polymerase will misincorporate a nucleotide into a growing DNA chain. These misincorporations need to be repaired prior to the next DNA replication or a mutation will occur.



3) The **3' to 5' proofreading exonuclease** activity of DNA polymerase clips off any unpaired residues at the primer terminus. It continues this activity until a base paired 3'-OH terminus is encountered.



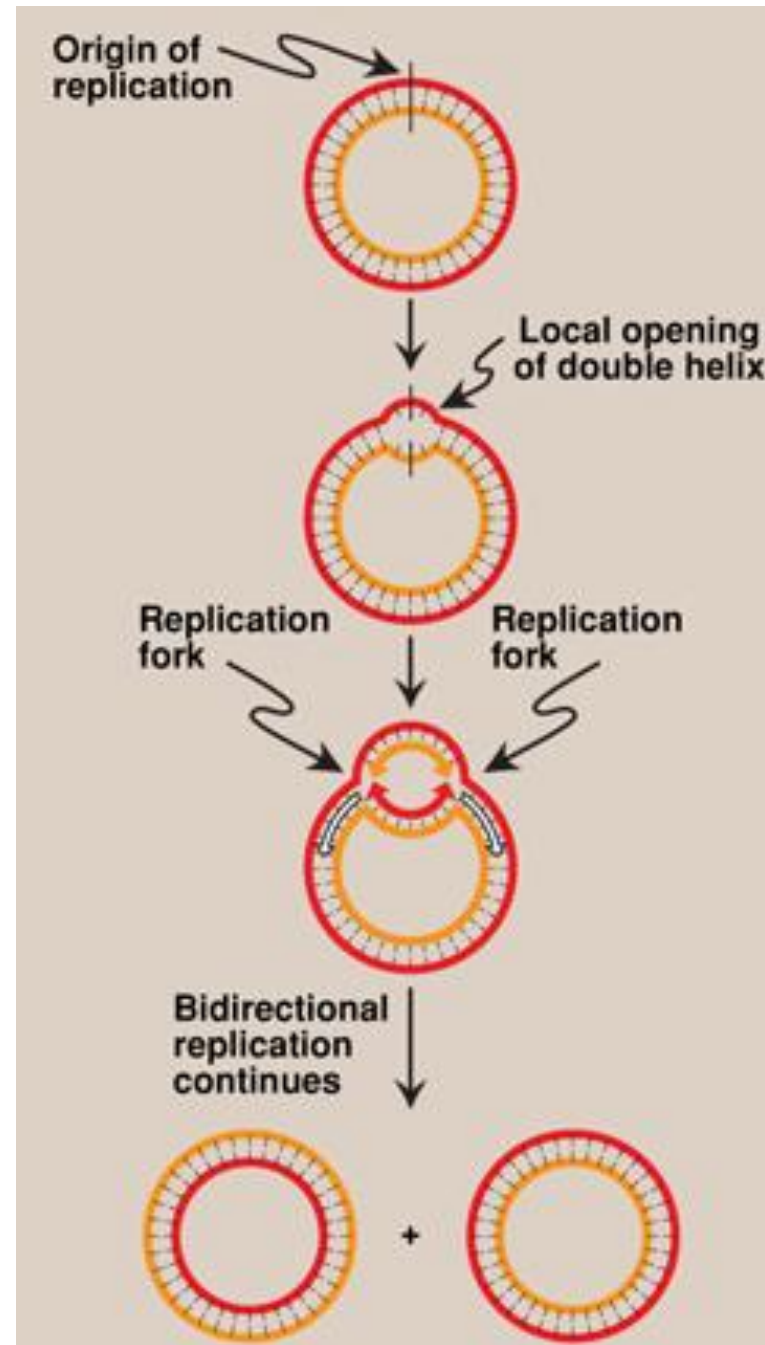
Bacterial genomes are usually circular and contain a single ORI (origin of replication) – **THETA REPLICATION**.

DNA replication is initiated at distinct sites called *origins of replication (ORI)*.

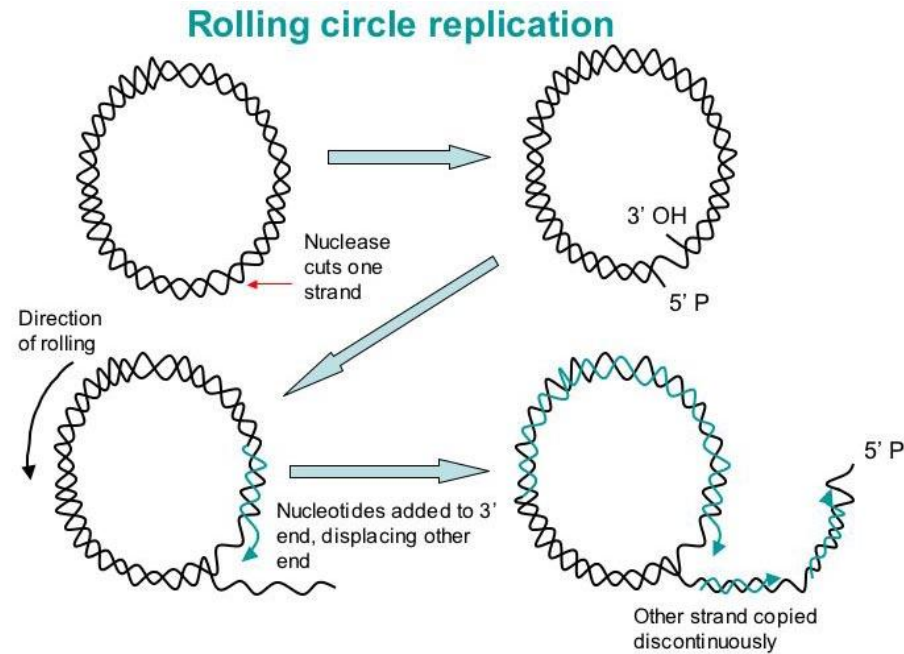
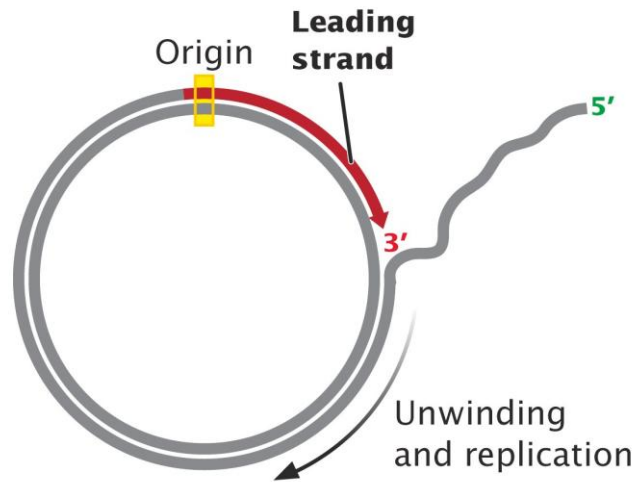
In *E. coli*, the replication process is only regulated at the point of initiation, therefore once the replication fork is established - replication proceeds until completion.

SOM.MK.I.BPM1.1.FTM.3.BCHM.0038

SOM.MK.I.BPM1.1.FTM.3.BCHM.0046



(b) Rolling-circle model



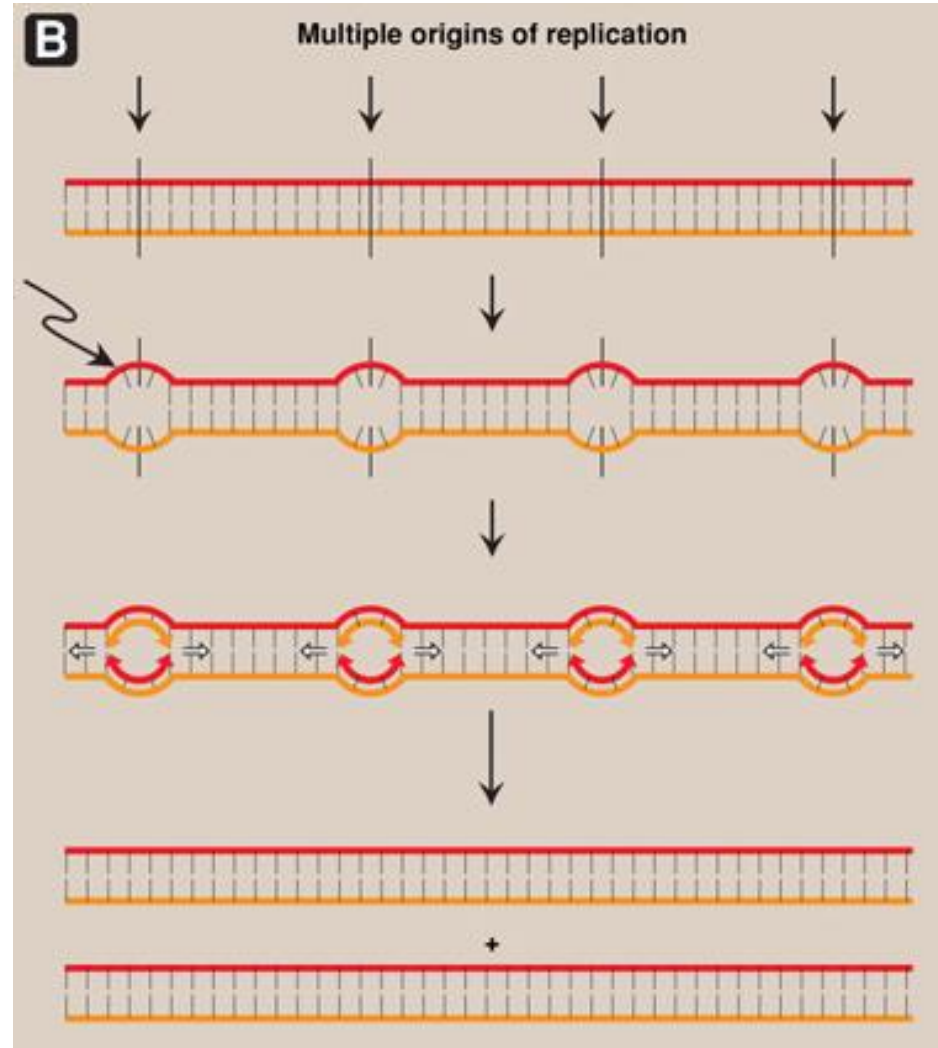
Single origin of replication

Various **bacterial virus (bacteriophage)** and **plasmids** that have a DNA genome utilize **ROLLING CIRCLE REPLICATION**.

SOM.MK.I.BPM1.1.FTM.3.BCHM.0038

SOM.MK.I.BPM1.1.FTM.3.BCHM.0046

Eukaryote
chromosomes have
many replication
origins and no
replication termini –
LINEAR
REPLICATION



SOM.MK.I.BPM1.1.FTM.3.BCHM.0038

SOM.MK.I.BPM1.1.FTM.3.BCHM.0046

Multiple origins of replication are required to replicate the large genomes that are found in eukaryotes in a timely fashion.

Human haploid genome = ~3.4 billion base pairs
Average size of chromosome = 100,000,000 bp

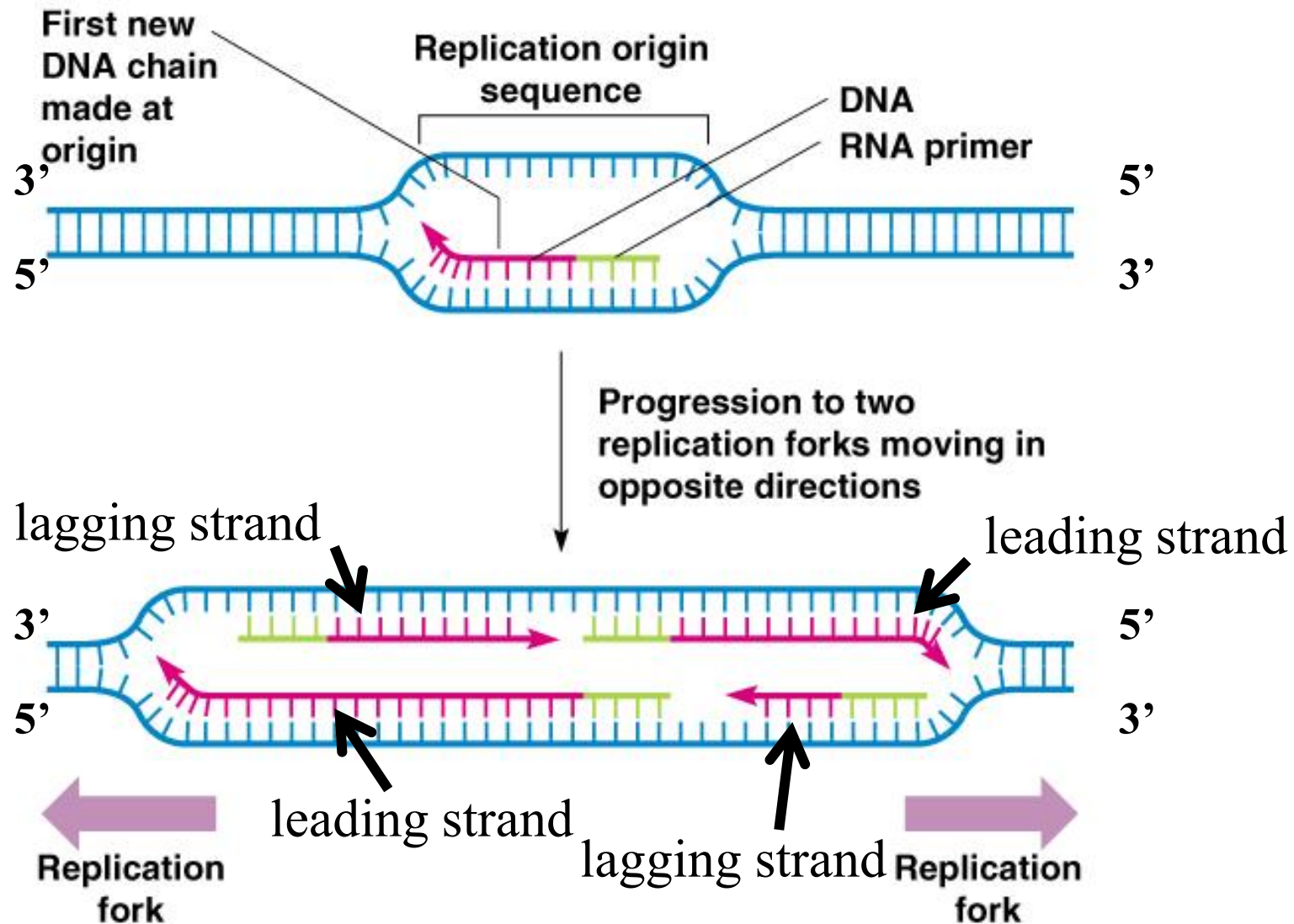
Rate of replication = 2000 bp/minute

Therefore, if there was a single origin of replication on each human chromosome it would take 830 hours for replication to complete.

<u>Organism</u>	<u>No. of replications</u>	<u>Fork Movement</u>
Bacteria	1	50,000 bp/min
Yeast	500	3,600 bp/min
Fruit fly	3,500	2,600 bp/min
Mouse	25,000	2,200 bp/min

The number of origins of replication increases in organisms containing large genomes.

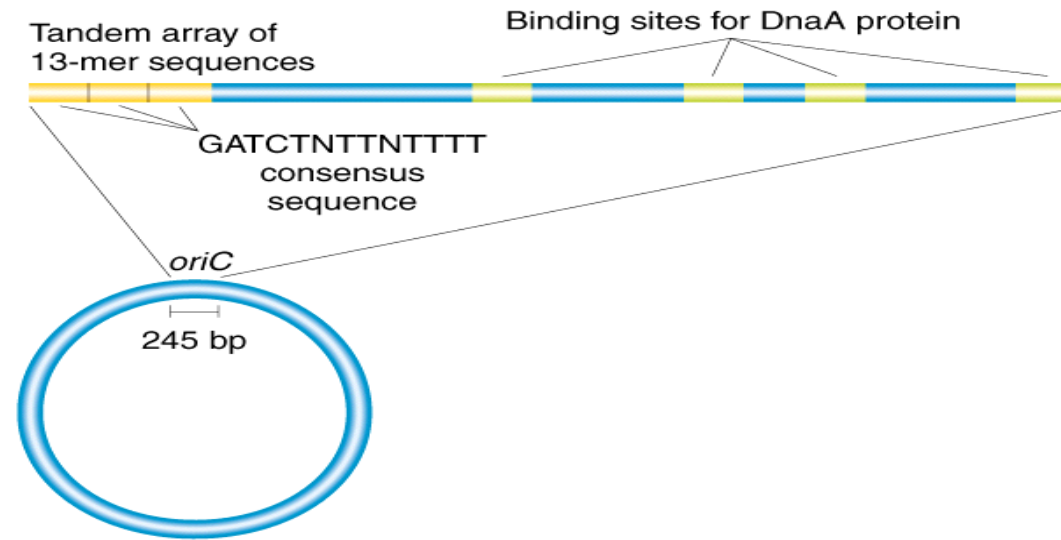
Diagram of the formation at a replication origin sequence of two replication forks that move in opposite directions



The Players: Initiation of DNA Replication (prokaryotes)

- 1) **Initiator proteins (DNaA protein)**: binds to origin of replication and breaks hydrogen bonds between bases.
- 2) **DNA helicase (DNaB)**: opens helix and binds primase to form primosome.
- 3) **Helicase inhibitor (DNaC)**: delivers helicase to DNA template
- 4) **DNA primase**: an RNA polymerase that synthesizes a RNA primer on the lagging strand to enable DNA polymerase to synthesize DNA strand.
- 5) **DNA polymerase I**: remove RNA primer and replace with DNA
- 6) **DNA polymerase III**: synthesis of leading and lagging strands
- 7) **Single stranded binding protein (ssb)**: binds to single stranded DNA in the replication bubble and prevents it from re-annealing or forming secondary structure.

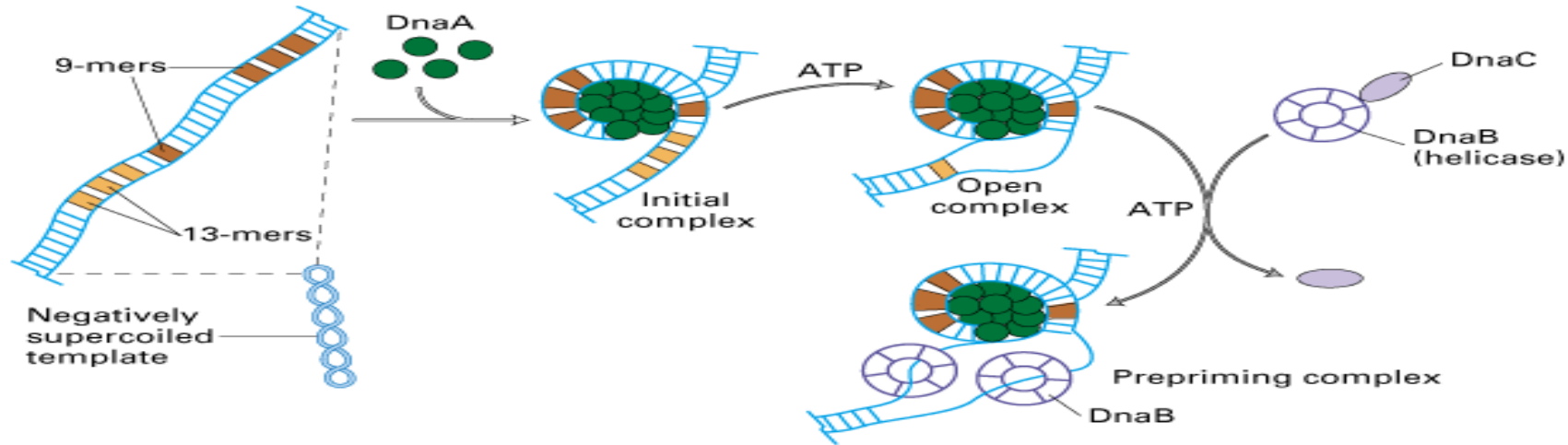
OriC, the origin of replication in *E.coli* has a length of 245 bp and contains a tandem array of three nearly identical 13-nucleotide sequences and four 9-nucleotide sequences that act as binding sites for DNA protein.



What is the significance of the DNA sequence at the *OriC*???

The *OriC* contains DNA sequences that are A-T rich. Bonding between A-T is the weakest compared to G-C bonding - thus the *OriC* facilitates easier melting and strand separation of the DNA molecule.

Model of initiation of replication at *E.coli* *OriC*

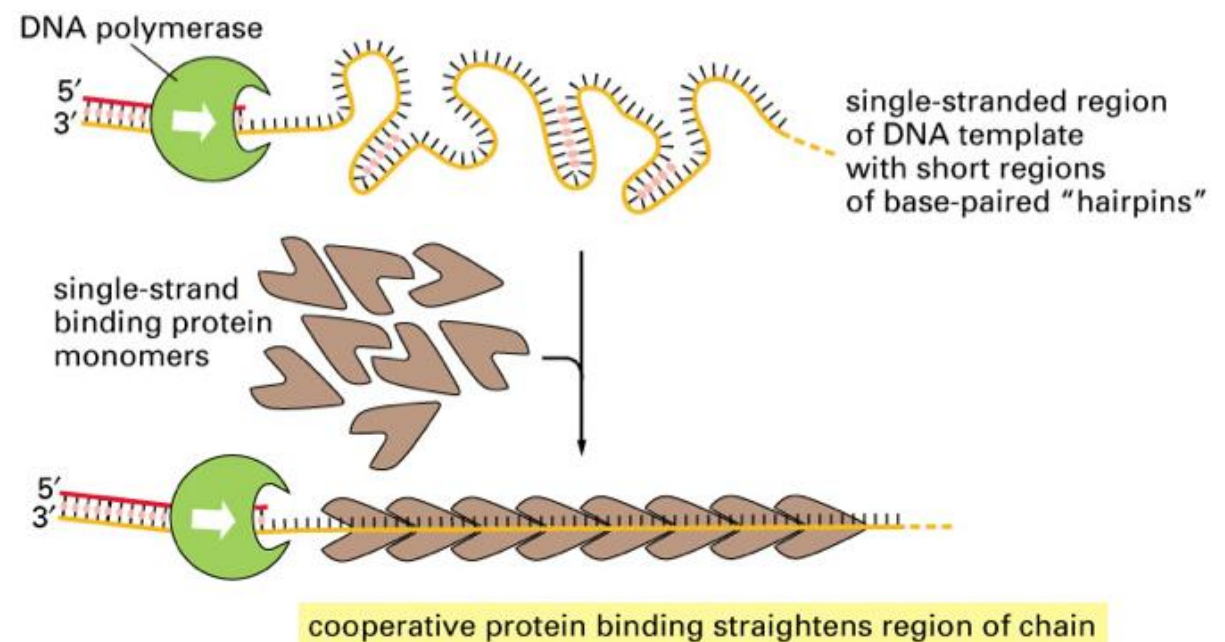


- 1) Multiple copies of Initiator proteins (DnaA) bind to the 9-mers at the origin
- 2) Strand separation occurs at the region of the 13-mer sequences
- 3) Helicase inhibitor (DnaC) protein delivers helicase (DnaB) to the template
- 4) Helicase clamps around each single strand of DNA
- 5) Helicase proceed to unwind the DNA in opposite directions away from origin

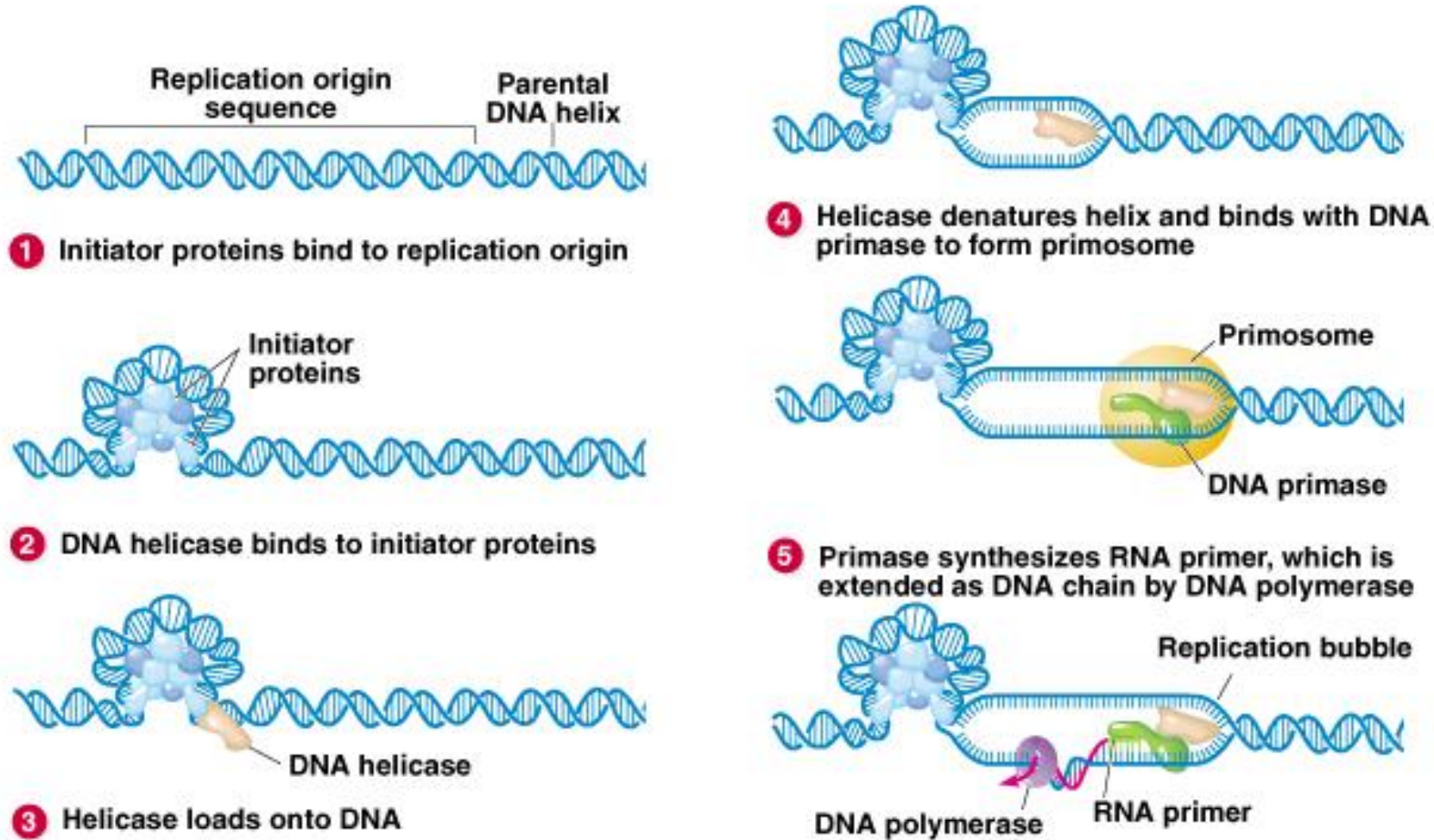
Problem #1: Single stranded DNA tends to re-anneal and form secondary structures.

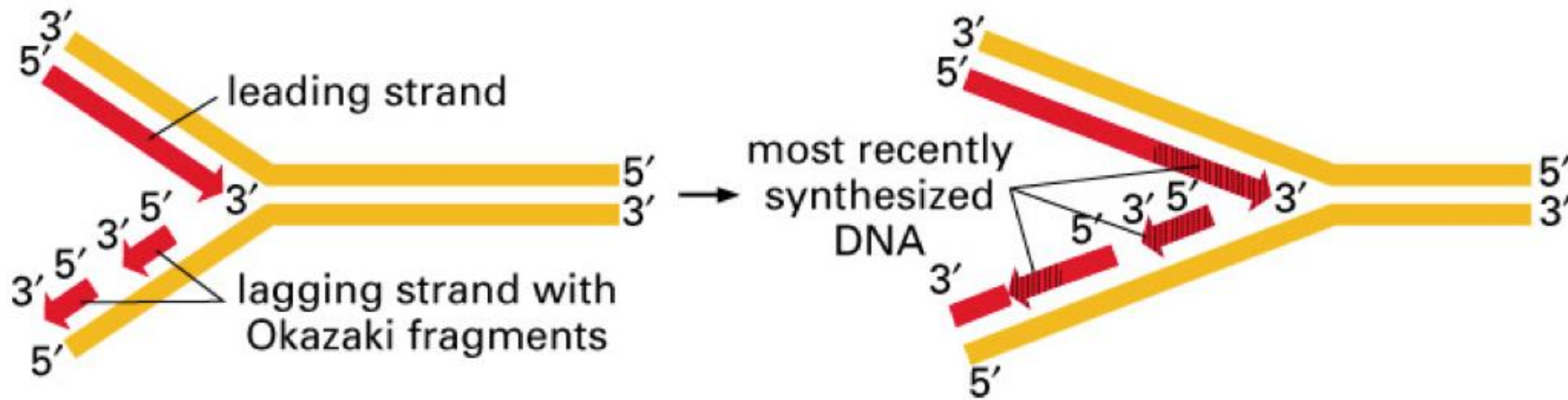
Solution: Single-strand binding proteins (SSB proteins).

SSB proteins binds to single stranded DNA in the replication fork. The cooperative binding of these proteins prevent reannealing and straightens out the DNA template to facilitate the DNA polymerization process.



Model for the formation of a replication bubble at a replication origin in *E. coli* and the initiation of the new DNA strand





DNA Synthesis on the lagging strand is discontinuous:

Daughter DNA strands are synthesized in the 5' to 3' direction, therefore the DNA synthesis on the lagging strand must be made initially as a series of short DNA molecules called **Okazaki fragments**.

Prokaryote DNA polymerases:

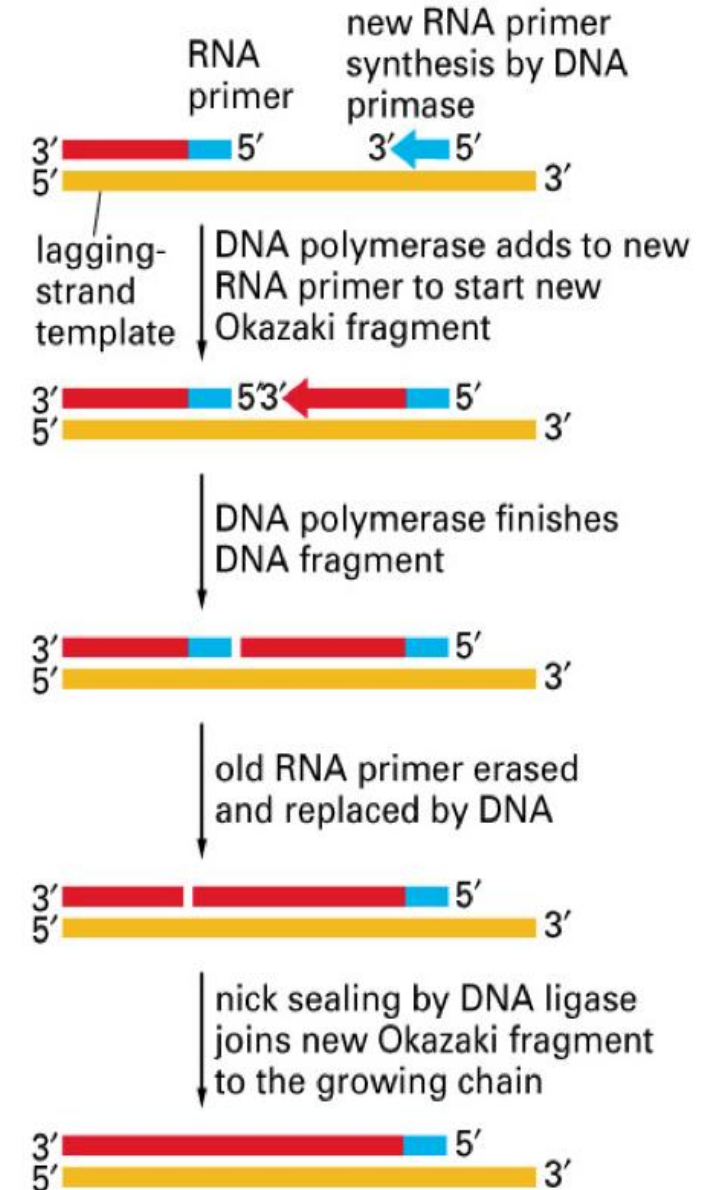
DNA polymerase III synthesizes the **Okazaki fragment** in addition to the leading strand.

DNA polymerase I removes the RNA primer (RNAseH and FEN-1 protein in eukaryotes).

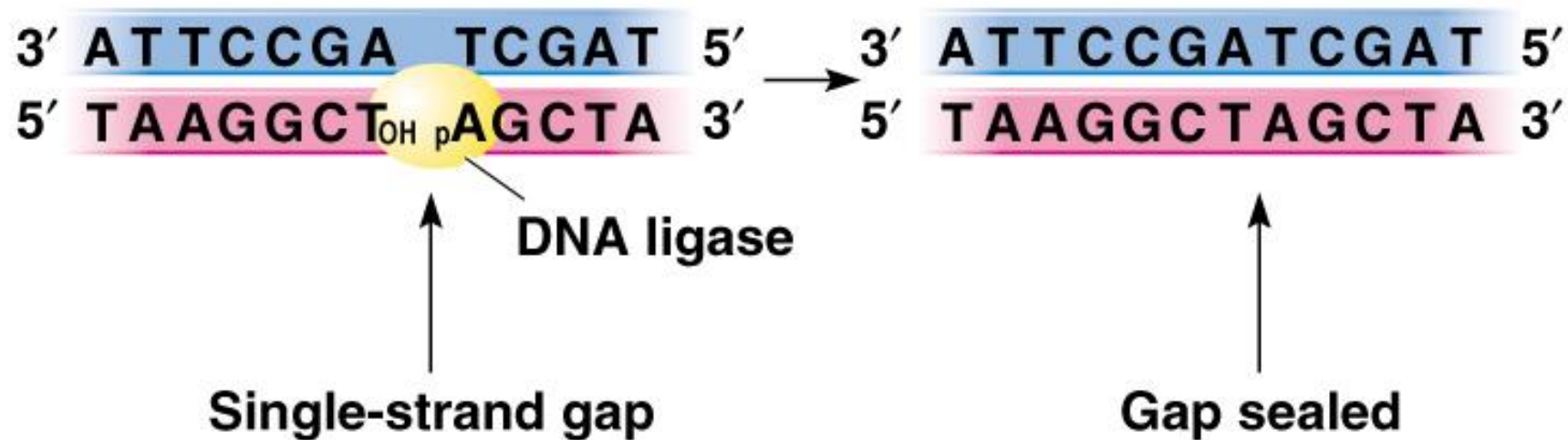
DNA polymerase I replaces the RNA primer with DNA

Problem #2: How are the Okazaki fragments joined together?

Solution: DNA ligase

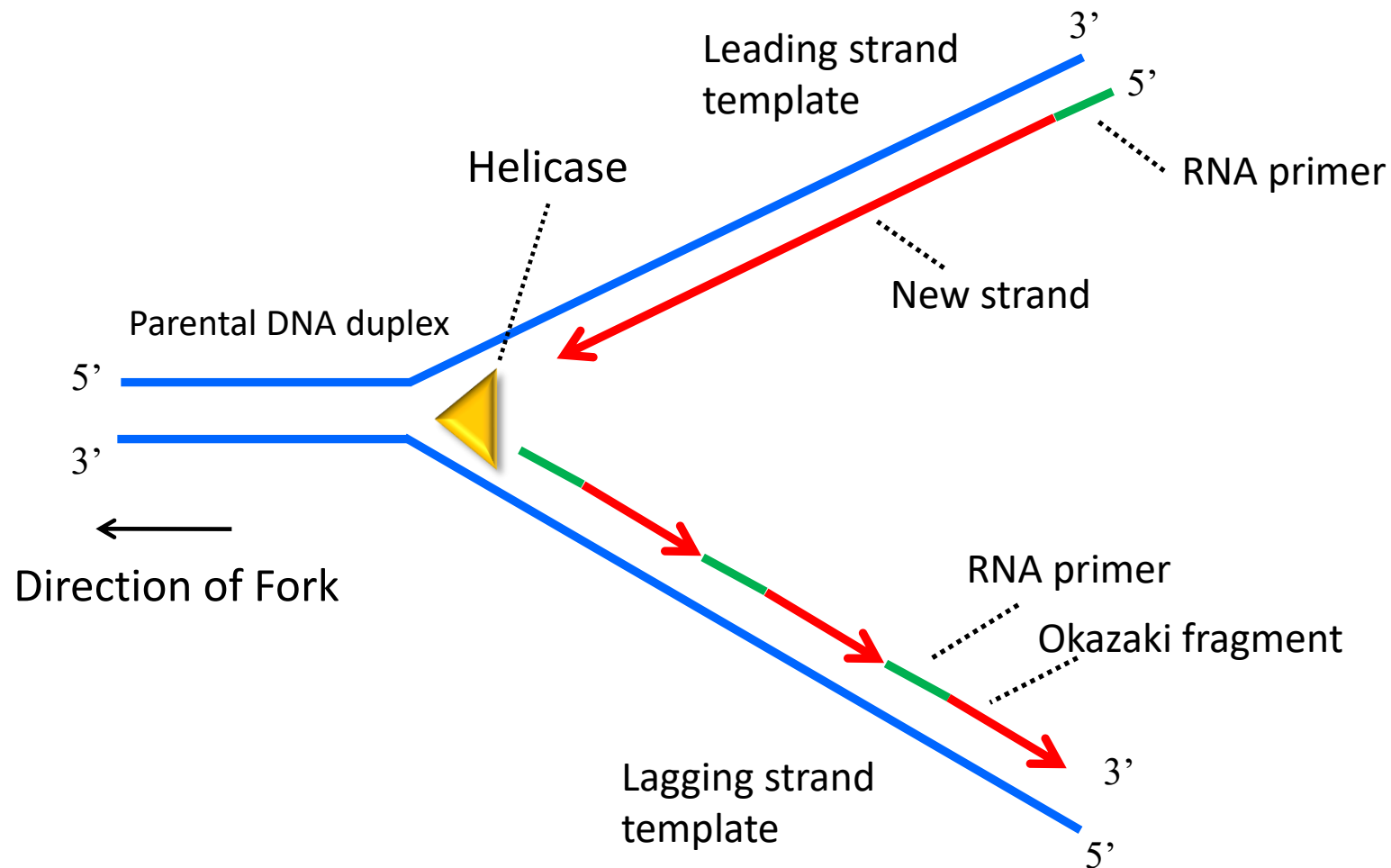


Action of DNA ligase in sealing the gap between adjacent DNA fragments to form a longer, covalently continuous chain.



Ligase utilizes ATP as the energy source for this reaction.

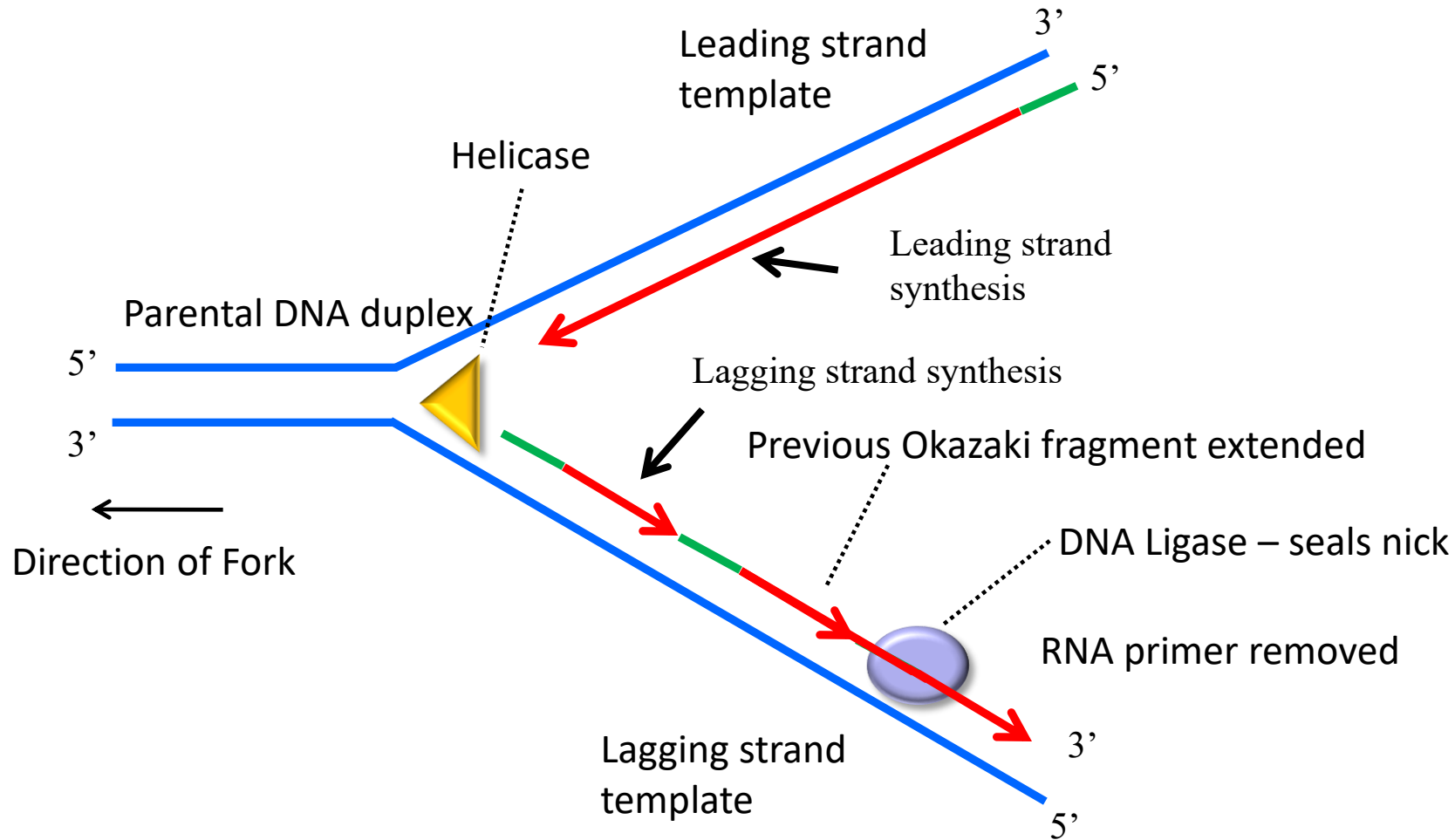
DNA polymerases synthesise new DNA in the 5' – 3' direction



Leading Strand synthesis - requires 1 priming event

Lagging strand synthesis - each Okazaki fragment requires a separate primer

DNA polymerases synthesise new DNA in the 5' – 3' direction

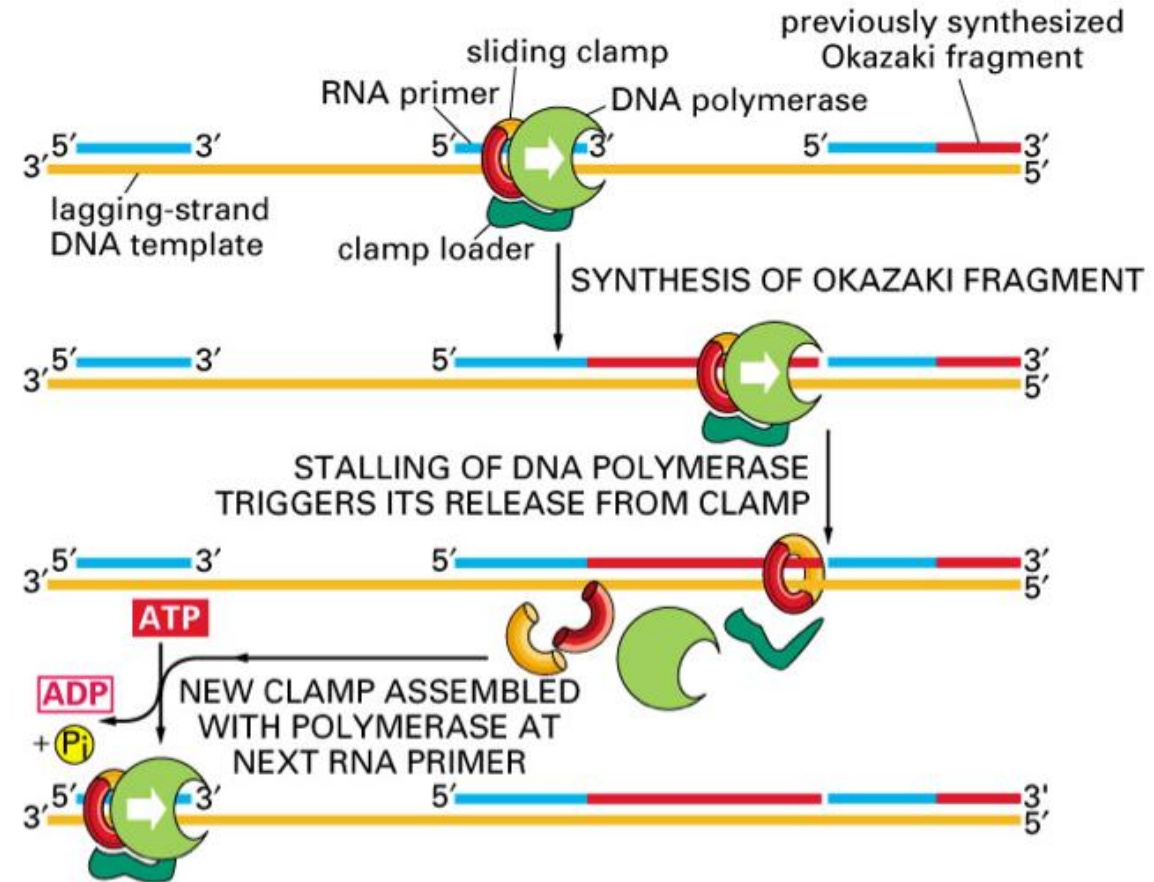


Following synthesis of Okazaki fragments:-

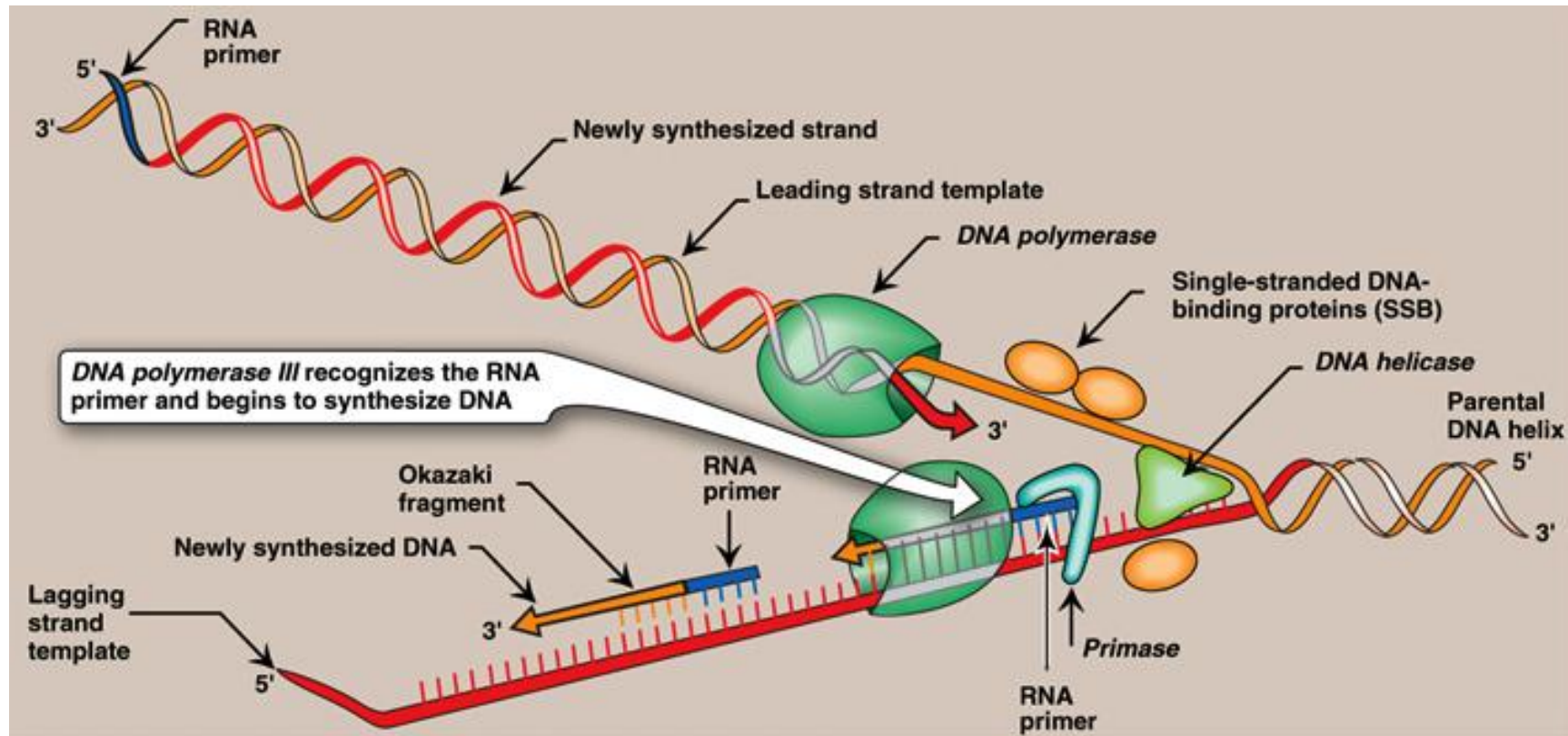
- DNA Pol I (prokaryotes) or RnaseH & FEN-1 protein (eukaryotes) removes the primer
- Gap filled by a DNA Pol
- Ligase joins the adjacent fragments

Problem #3: How is DNA polymerase III loaded and maintained on the single stranded DNA template, and unloaded when it reaches double stranded DNA (eg. The next Okazaki fragment).

Solution: Clamp protein
Clamp protein tightly holds the DNA polymerase onto the template for synthesis of long templates (increases expressivity), and releases DNA pol when it stalls at a region of double stranded DNA.



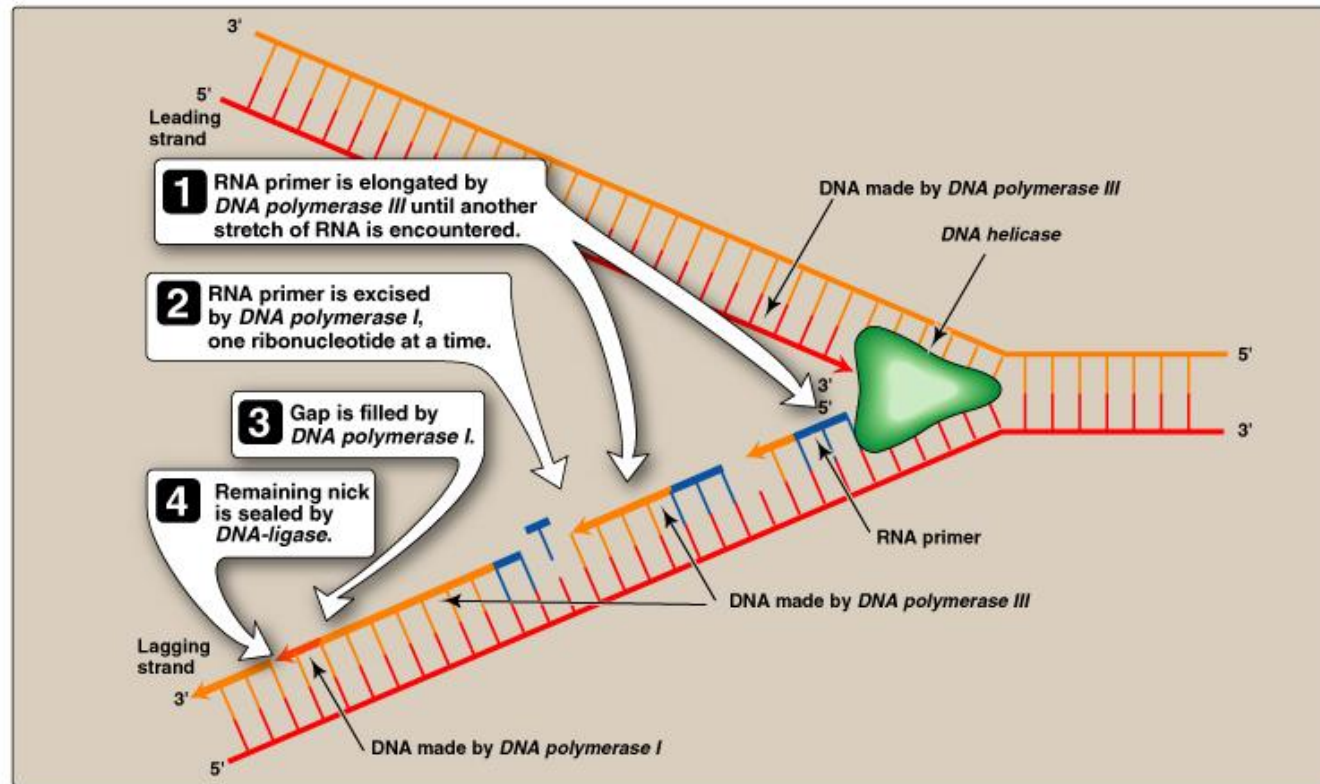
Summary of DNA Replication in *E.coli*



Summary of DNA Replication in *E.coli*

Gap Filling by DNA Pol I

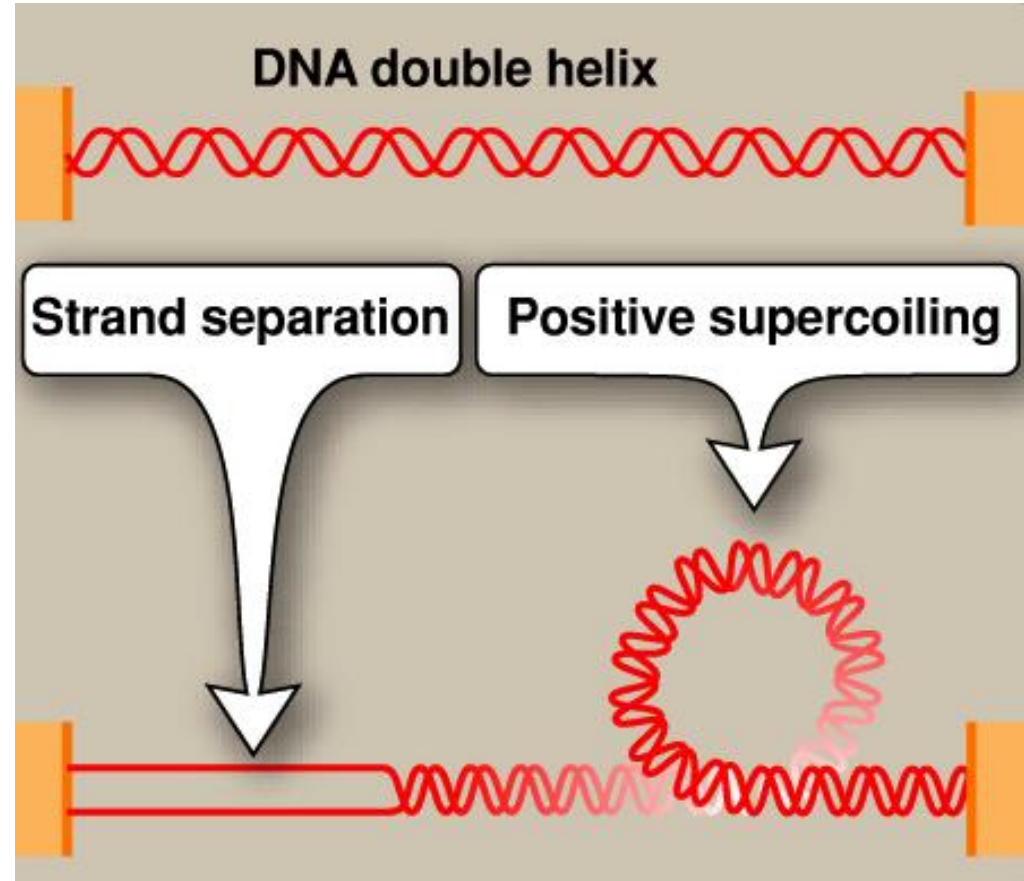
5'- 3' Polymerase activity allows Pol I to add dNTPs to the 3' end of the previous Okazaki fragment



When the gap is filled, Pol I dissociates leaving a nick. DNA ligase binds to the nick & catalyses the formation of a phosphodiester bond

Problem #4: For a bacterial replication fork moving at 500 bp per second, the parental DNA helix ahead of the fork must rotate at 50 revolutions per minute! This generates positive supercoils ahead of the replication fork.

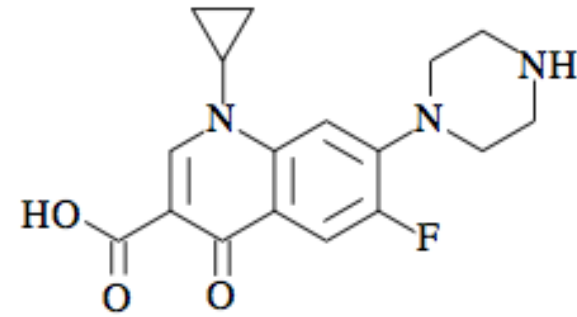
Solution: Swivel the DNA using **Topoisomerase enzymes**



DNA gyrase (type II topoisomerase):

This enzyme introduces negative supercoils into the DNA. This reduces the positive supercoils introduced by the opening of the DNA.

It also aids in the separation of the DNA during replication and transcription.



Ciprofloxacin

Ciprofloxacin is a quinolone drug that inhibits bacterial DNA gyrase.

It is used in the treatment of respiratory and urinary tract infections and can be used to treat anthrax.

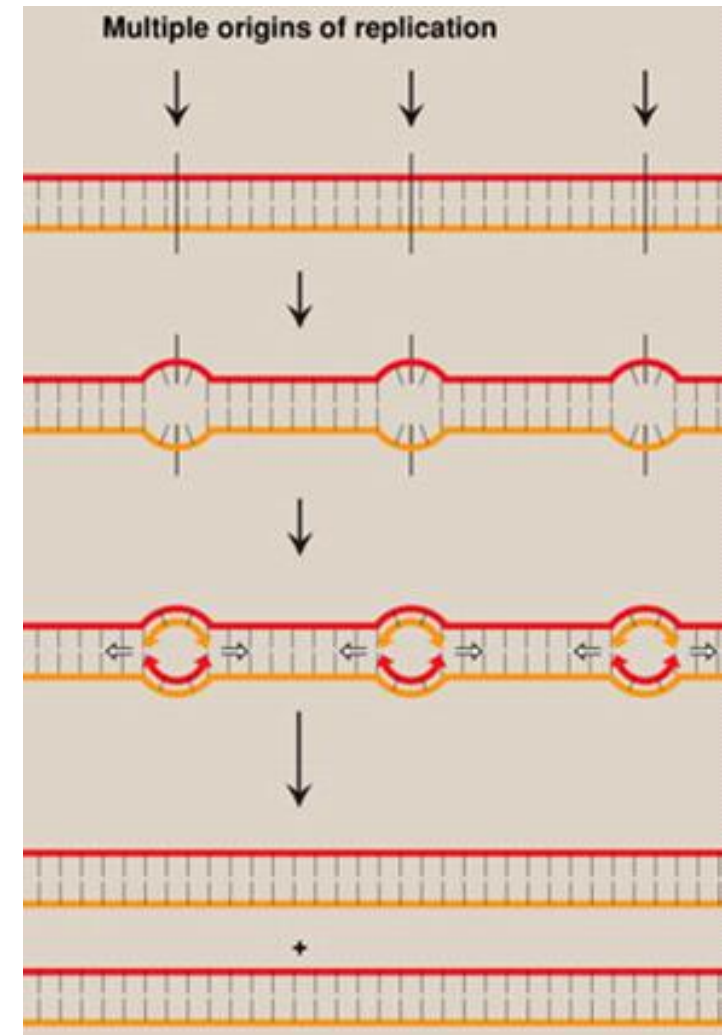
DNA replication in Eukaryotes

Mechanism shows strong parallels but more complex!

Major differences:

1. Origins of replication
2. Polymerases
3. Removal of primers
4. Compartmentalized into the S phase
5. Telomerase and the replication of the ends of eukaryote chromosomes
(DLA study assignment)****

Depending on the organism & cell type, replication origins occur every 3 – 300 kbp (an average human chromosome has several hundred replication origins).



Comparison between prokaryote and eukaryote DNA replication:

1. Single origins of replication in prokaryotes, multiple origins of replication in eukaryotes.
2. RNA primer in prokaryotes is removed by DNA polymerase I, RNA primer is removed by RNAaseH and FEN1.
3. Different DNA polymerases:

Prokaryotic DNA polymerases:

DNA polymerase I – removal of RNA primer and replacement with DNA

DNA polymerase III – synthesis of leading and lagging strands
- Main replication enzyme

Eukaryotic DNA polymerases:

Polymerase- α :

- Synthesizes RNA primer on leading and lagging strand
- Subunit possesses primase activity and initiates DNA synthesis
- Low processivity – falls off after synthesizing a short DNA segment
- No exonuclease activity

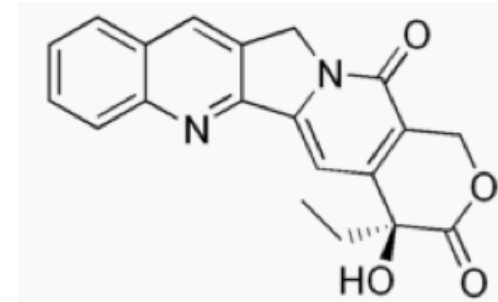
Polymerase- δ :

- Synthesizes DNA from leading and lagging strands
- Main replicative enzyme
- High processivity – can synthesize long stretches of DNA
- 3'-5' exonuclease proofreading activity

Other polymerases – β and γ are involved in DNA repair and mitochondria genome replication, respectively.

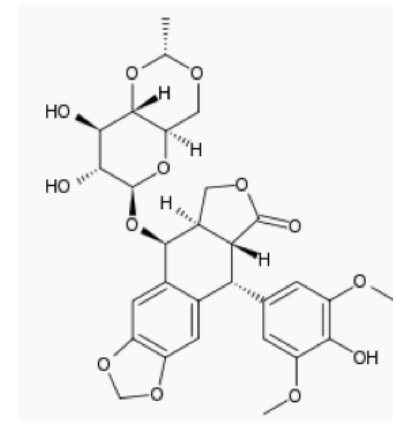
Inhibitors of eukaryote replication

Camptothecin, an anticancer drug, binds to and inhibits topoisomerase I activity. The result is DNA breakage.



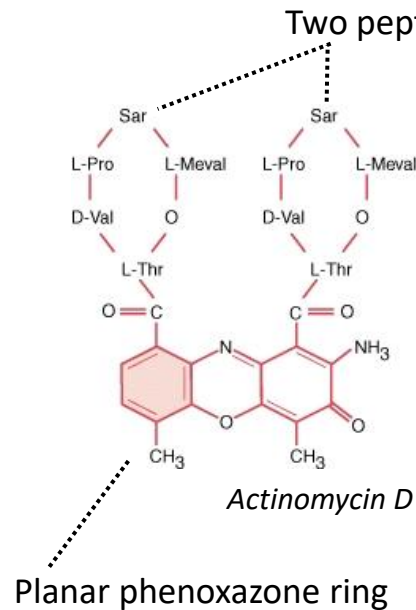
Camptothecin

Etoposide, another anticancer drug, inhibits the activity of topoisomerase II.



Etoposide

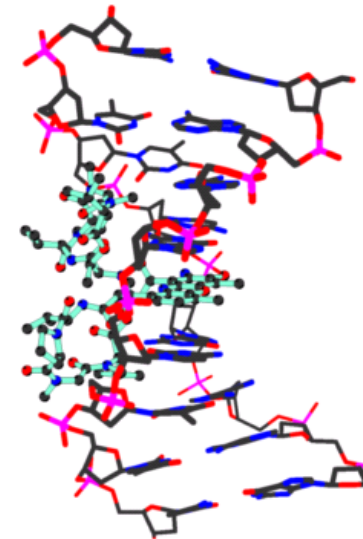
Inhibitors of replication – Actinomycin D (Dactinomycin)



The phenoxazone ring intercalates
between adjacent guanine-cytosine bases

→

The polypeptide chains extend
along the minor groove of the
helix, thereby stabilizing the drug-
DNA complex.



Prevents DNA replication by:

- ❖ Preventing the formation of regions of ss DNA.
- ❖ Also, inhibits RNA transcription (as it prevents strand separation)